Skills Summary

Modeling Periodic Behavior from Graphs

Use this reference while completing your worksheet or when studying.

Skill 1 — Amplitude and midline from a graph

From the two extremes (highest value h, lowest value l):

amplitude $A = \dfrac{h-l}{2}$ (half the spread) midline $y = D$ with $D = \dfrac{h+l}{2}$ (the average).

Amplitude is measured *from the midline*, not from the horizontal axis.

Example: A graph rises to 7 and falls to -3. Find the amplitude and the midline.
Step 1. The two extremes are all that is needed — half their difference is the amplitude, their average is the midline.
Step 2. $A = \dfrac{7-(-3)}{2} = \dfrac{10}{2}$ and $D = \dfrac{7+(-3)}{2} = \dfrac{4}{2} \Rightarrow$ $A = \mathbf{5}$, midline $y = \mathbf{2}$.

Try it: A graph rises to 11 and falls to 5. Find the amplitude and the midline value.

check: $A = \mathbf{3}$, midline $y = \mathbf{8}$

Skill 2 — Period and the value of B

Period is the horizontal length of one complete cycle: peak to next peak, or crossing to the matching crossing.

For $y = A\sin(Bx) + D$: period $= \dfrac{2\pi}{B}$, so $B = \dfrac{2\pi}{\text{period}}$.

Period longer than 2π gives $B < 1$; period shorter gives $B > 1$.

Example: A graph repeats every 6 units. Find B.
Step 1. The picture gives the period directly, and B is what turns a period into the standard 2π cycle.
Step 2. $B = \dfrac{2\pi}{6} \Rightarrow$ $B = \boldsymbol{\frac{\pi}{3}}$

Try it: A graph repeats every 10 units. Find B.

check: $B = \boldsymbol{\frac{\pi}{5}}$

Skill 3 — Writing the model from a graph

Three readings, three parameters. Highest h, lowest l, period p:

$A = \dfrac{h-l}{2}$, $D = \dfrac{h+l}{2}$, $B = \dfrac{2\pi}{p}$.

Then pick the shape: a peak at $x = 0$ takes cos; a midline crossing on the way up takes sin.

Example: A graph peaks at 12, dips to 4, repeats every 4 units, and peaks at $x = 0$. Write the model.

Step 1. A peak at $x = 0$ is cosine's own starting shape, so no shift is needed and only A, D and B remain.

Step 2. $A = \dfrac{12-4}{2}$, $D = \dfrac{12+4}{2}$, $B = \dfrac{2\pi}{4} \Rightarrow$ $A = \mathbf{4}$, $D = \mathbf{8}$, $B = \boldsymbol{\frac{\pi}{2}}$, so $y = 4\cos\left(\frac{\pi}{2}x\right) + 8$.

Try it: A graph peaks at 30, dips to 10, repeats every 12 units, and peaks at $x = 0$. Find A, D and B.

check: $A = \mathbf{10}$, $D = \mathbf{20}$, $B = \boldsymbol{\frac{\pi}{6}}$

Skill 4 — Solving with the model

Method: isolate the sine or cosine, then use *both* angles in one turn that have that value — for sine, θ and $\pi - \theta$; for cosine, θ and $-\theta$. Undo the inside last.

A full period of the input always yields two solutions unless the target is an extreme value.

Example: Solve $4\sin\left(\dfrac{\pi t}{6}\right) + 12 = 14$ for $0 \le t < 12$.

Step 1. Strip the outside first so the equation reads sine equals a number, then take both angles with that sine.

Step 2. $\sin\left(\dfrac{\pi t}{6}\right) = \dfrac{1}{2} \Rightarrow \dfrac{\pi t}{6} = \dfrac{\pi}{6}$ or $\dfrac{5\pi}{6} \Rightarrow$ $t = \mathbf{1}$ or $t = \mathbf{5}$.

Try it: Solve $6\sin\left(\dfrac{\pi t}{4}\right) + 10 = 13$ for $0 \le t < 8$.

check: $t = \boldsymbol{\frac{2}{3}}$ or $t = \boldsymbol{\frac{10}{3}}$